In [33]:
```
#importing  python libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as mpl
import sklearn
```

In [34]:
```
df=pd.read_csv("healthcare_dataset.csv.zip")
```

In [35]:
```
df.head(10)
```

Out[35]:

| | Name | Age | Gender | Blood Type | Medical Condition | Date of Admission | Doctor | Hospital | I |
|---|---|---|---|---|---|---|---|---|---|
| **0** | Bobby JacksOn | 30 | Male | B- | Cancer | 2024-01-31 | Matthew Smith | Sons and Miller | B |
| **1** | LesLie TErRy | 62 | Male | A+ | Obesity | 2019-08-20 | Samantha Davies | Kim Inc | |
| **2** | DaNnY sMitH | 76 | Female | A- | Obesity | 2022-09-22 | Tiffany Mitchell | Cook PLC | |
| **3** | andrEw waTtS | 28 | Female | O+ | Diabetes | 2020-11-18 | Kevin Wells | Hernandez Rogers and Vang, | |
| **4** | adrIENNE bEll | 43 | Female | AB+ | Cancer | 2022-09-19 | Kathleen Hanna | White-White | |
| **5** | EMILY JOHNSOn | 36 | Male | A+ | Asthma | 2023-12-20 | Taylor Newton | Nunez-Humphrey | UnitedH |
| **6** | edwArD EDWaRDs | 21 | Female | AB- | Diabetes | 2020-11-03 | Kelly Olson | Group Middleton | |
| **7** | CHrisTInA MARtinez | 20 | Female | A+ | Cancer | 2021-12-28 | Suzanne Thomas | Powell Robinson and Valdez, | |
| **8** | JASmINe aGuIlaR | 82 | Male | AB+ | Asthma | 2020-07-01 | Daniel Ferguson | Sons Rich and | |
| **9** | ChRISTopher BerG | 58 | Female | AB- | Cancer | 2021-05-23 | Heather Day | Padilla-Walker | UnitedH |

In [36]:
```
#Checking for the number of rows and columns
df.shape
```

Out[36]:
```
(55500, 15)
```

In [37]:
```
#Checking if there are any duplicates
df.duplicated().any()
```

Out[37]: np.True_

In [38]:
```
#Checking for the number of duplicate values
df.duplicated().sum()
```

Out[38]: np.int64(534)

In [39]:
```
#Viewing the duplicated rows
duplicated_view=df[df.duplicated(keep=False)]
print(duplicated_view.head(10))
```

```
                Name  Age  Gender Blood Type Medical Condition  \
64      Nancy glOVeR   58    Male         A-      Hypertension
107    DAVid higgInS   49  Female         B-         Arthritis
148   RoBErt hIGGInS   42    Male        AB-            Asthma
154      kevIn HiCKs   66    Male        AB+         Arthritis
159   miCHAeL TayloR   29    Male         O-            Asthma
246    wiLlIAm mcCOY   80    Male         A-            Asthma
360    tHOMas DENnIs   40    Male         A+            Asthma
408  MelIssA hOFfMAN   36    Male         B+            Cancer
454      jOrdan LEON   51  Female         O+         Arthritis
621   JesSIcA caRtER   51  Female         A-           Obesity

    Date of Admission             Doctor                     Hospital  \
64         2020-05-08    Jennifer Larson  Khan, and Rodriguez Fischer
107        2021-03-05  Erin Henderson MD    Evans and Hall Schneider,
148        2021-05-06        Scott Davis      and Ford Lee, Rodriguez
154        2021-06-23       Kelly Murphy                Robinson Inc
159        2020-02-27     Erica Mccormick              Donaldson-Frey
246        2023-02-20     Whitney Vaughan       and Walker Glenn, Long
360        2024-01-28        Leonard Chen             Mckinney-Edwards
408        2022-07-27      April Cummings                  Cooley-Reid
454        2023-08-11     Christina Lopez                 Dennis-Roth
621        2019-12-08        Sarah Ibarra              Anderson Group

    Insurance Provider  Billing Amount  Room Number Admission Type  \
64            Medicare    19183.168885          378      Emergency
107           Medicare    24948.477824          361      Emergency
148           Medicare    13355.782085          451       Elective
154           Medicare     1897.891727          196       Elective
159           Medicare    41939.119937          453       Elective
246              Cigna     3845.397496          158      Emergency
360           Medicare    42959.299814          410       Elective
408              Cigna    41311.293901          434      Emergency
454              Cigna    21534.715570          198      Emergency
621              Cigna    32006.031566          200         Urgent

    Discharge Date   Medication  Test Results
64      2020-06-01      Aspirin      Abnormal
107     2021-03-20   Penicillin      Abnormal
148     2021-05-29    Ibuprofen  Inconclusive
154     2021-07-09    Ibuprofen      Abnormal
159     2020-03-26    Ibuprofen        Normal
246     2023-03-10   Penicillin        Normal
360     2024-02-21  Paracetamol      Abnormal
408     2022-08-19      Lipitor  Inconclusive
454     2023-08-20   Penicillin        Normal
621     2020-01-05   Penicillin        Normal
```

In [40]:
```
#Dropping the duplicate values
df=df.drop_duplicates()
print(df)
```

```
                    Name  Age  Gender Blood Type Medical Condition  \
0          Bobby JacksOn   30    Male         B-            Cancer
1           LesLie TErRy   62    Male         A+           Obesity
2            DaNnY sMitH   76  Female         A-           Obesity
3           andrEw waTtS   28  Female         O+          Diabetes
4          adrIENNE bEll   43  Female        AB+            Cancer
...                  ...  ...     ...        ...               ...
55495  eLIZABeTH jaCkSOn   42  Female         O+            Asthma
55496         KYle pEREz   61  Female        AB-           Obesity
55497       HEATher WaNG   38  Female         B+      Hypertension
55498     JENniFER JOneS   43    Male         O-         Arthritis
55499       jAMES GARCiA   53  Female         O+         Arthritis

      Date of Admission            Doctor                     Hospital  \
0            2024-01-31     Matthew Smith              Sons and Miller
1            2019-08-20   Samantha Davies                      Kim Inc
2            2022-09-22  Tiffany Mitchell                     Cook PLC
3            2020-11-18       Kevin Wells    Hernandez Rogers and Vang,
4            2022-09-19    Kathleen Hanna                  White-White
...                 ...               ...                          ...
55495        2020-08-16     Joshua Jarvis               Jones-Thompson
55496        2020-01-23   Taylor Sullivan                 Tucker-Moyer
55497        2020-07-13    Joe Jacobs DVM  and Mahoney Johnson Vasquez,
55498        2019-05-25    Kimberly Curry      Jackson Todd and Castro,
55499        2024-04-02     Dennis Warren               Henry Sons and

      Insurance Provider  Billing Amount  Room Number Admission Type  \
0            Blue Cross    18856.281306          328         Urgent
1              Medicare    33643.327287          265      Emergency
2                 Aetna    27955.096079          205      Emergency
3              Medicare    37909.782410          450       Elective
4                 Aetna    14238.317814          458         Urgent
...                 ...             ...          ...            ...
55495        Blue Cross     2650.714952          417       Elective
55496             Cigna    31457.797307          316       Elective
55497   UnitedHealthcare    27620.764717          347         Urgent
55498          Medicare    32451.092358          321       Elective
55499             Aetna     4010.134172          448         Urgent

      Discharge Date   Medication  Test Results
0         2024-02-02  Paracetamol        Normal
1         2019-08-26    Ibuprofen  Inconclusive
2         2022-10-07      Aspirin        Normal
3         2020-12-18    Ibuprofen      Abnormal
4         2022-10-09   Penicillin      Abnormal
...              ...          ...           ...
55495     2020-09-15   Penicillin      Abnormal
55496     2020-02-01      Aspirin        Normal
55497     2020-08-10    Ibuprofen      Abnormal
55498     2019-05-31    Ibuprofen      Abnormal
55499     2024-04-29    Ibuprofen      Abnormal

[54966 rows x 15 columns]
```

In [41]:
```
#Making sure dupicated values have been dropped
print(f"remaining duplicate rows:{df.duplicated().sum()}")
```

```
remaining duplicate rows:0
```

In [42]:
```
#Checking if there are missing values
df.isnull().any()
```

Out[42]:
```
Name                  False
Age                   False
Gender                False
Blood Type            False
Medical Condition     False
Date of Admission     False
Doctor                False
Hospital              False
Insurance Provider    False
Billing Amount        False
Room Number           False
Admission Type        False
Discharge Date        False
Medication            False
Test Results          False
dtype: bool
```

In [43]:
```
#Checking for the number of values for each column
df.isnull().sum()
```

Out[43]:
```
Name                  0
Age                   0
Gender                0
Blood Type            0
Medical Condition     0
Date of Admission     0
Doctor                0
Hospital              0
Insurance Provider    0
Billing Amount        0
Room Number           0
Admission Type        0
Discharge Date        0
Medication            0
Test Results          0
dtype: int64
```

In [44]:
```
#Checking for the data types
df.dtypes
```

Out[44]:
```
Name                   object
Age                     int64
Gender                 object
Blood Type             object
Medical Condition      object
Date of Admission      object
Doctor                 object
Hospital               object
Insurance Provider     object
Billing Amount        float64
Room Number             int64
Admission Type         object
Discharge Date         object
Medication             object
Test Results           object
dtype: object
```

In [45]:
```
#Converting the Date of Admission and Discharge Date columns from text data types f
df["Date of Admission"] = pd.to_datetime(df["Date of Admission"])
df["Discharge Date"] = pd.to_datetime(df["Discharge Date"])
```

In [46]:
```
#Confirming the conversion of the data types have occurred
df.dtypes
```

Out[46]:
```
Name                          object
Age                            int64
Gender                        object
Blood Type                    object
Medical Condition             object
Date of Admission     datetime64[ns]
Doctor                        object
Hospital                      object
Insurance Provider            object
Billing Amount               float64
Room Number                    int64
Admission Type                object
Discharge Date        datetime64[ns]
Medication                    object
Test Results                  object
dtype: object
```

In [47]:
```
#Correcting errors like "dANny smITH" to "Danny Smith" in the Name column
df["Name"] = df["Name"].str.strip().str.title()
```

In [48]:
```
df.head(10)
```

Out[48]:

| | Name | Age | Gender | Blood Type | Medical Condition | Date of Admission | Doctor | Hospital | In F |
|---|---|---|---|---|---|---|---|---|---|
| **0** | Bobby Jackson | 30 | Male | B- | Cancer | 2024-01-31 | Matthew Smith | Sons and Miller | Bl |
| **1** | Leslie Terry | 62 | Male | A+ | Obesity | 2019-08-20 | Samantha Davies | Kim Inc | N |
| **2** | Danny Smith | 76 | Female | A- | Obesity | 2022-09-22 | Tiffany Mitchell | Cook PLC | |
| **3** | Andrew Watts | 28 | Female | O+ | Diabetes | 2020-11-18 | Kevin Wells | Hernandez Rogers and Vang, | N |
| **4** | Adrienne Bell | 43 | Female | AB+ | Cancer | 2022-09-19 | Kathleen Hanna | White-White | |
| **5** | Emily Johnson | 36 | Male | A+ | Asthma | 2023-12-20 | Taylor Newton | Nunez-Humphrey | UnitedHe |
| **6** | Edward Edwards | 21 | Female | AB- | Diabetes | 2020-11-03 | Kelly Olson | Group Middleton | N |
| **7** | Christina Martinez | 20 | Female | A+ | Cancer | 2021-12-28 | Suzanne Thomas | Powell Robinson and Valdez, | |
| **8** | Jasmine Aguilar | 82 | Male | AB+ | Asthma | 2020-07-01 | Daniel Ferguson | Sons Rich and | |
| **9** | Christopher Berg | 58 | Female | AB- | Cancer | 2021-05-23 | Heather Day | Padilla-Walker | UnitedHe |

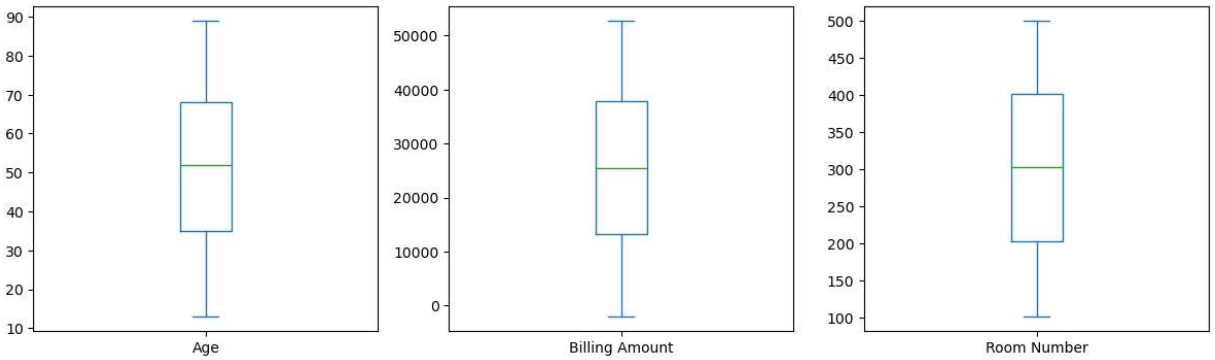
In [49]:
```
#Descriptive Statistics for the dataset
df.describe()
```

Out[49]:

| | Age | Date of Admission | Billing Amount | Room Number | Discharge Date |
|---|---|---|---|---|---|
| **count** | 54966.000000 | 54966 | 54966.000000 | 54966.000000 | 54966 |
| **mean** | 51.535185 | 2021-11-01 17:35:29.505512448 | 25544.306284 | 301.124404 | 2021-11-17 05:34:28.202161408 |
| **min** | 13.000000 | 2019-05-08 00:00:00 | -2008.492140 | 101.000000 | 2019-05-09 00:00:00 |
| **25%** | 35.000000 | 2020-07-28 00:00:00 | 13243.718641 | 202.000000 | 2020-08-13 00:00:00 |
| **50%** | 52.000000 | 2021-11-02 00:00:00 | 25542.749145 | 302.000000 | 2021-11-18 00:00:00 |
| **75%** | 68.000000 | 2023-02-03 00:00:00 | 37819.858159 | 401.000000 | 2023-02-19 00:00:00 |
| **max** | 89.000000 | 2024-05-07 00:00:00 | 52764.276736 | 500.000000 | 2024-06-06 00:00:00 |
| **std** | 19.605661 | NaN | 14208.409711 | 115.223143 | NaN |

In [50]:
```
#Checking for outliers
num_cols = df.select_dtypes(include="number").columns
import matplotlib.pyplot as plt

df[num_cols].plot(kind="box", subplots=True, layout=(2, 3), figsize=(12, 7))
plt.tight_layout()
plt.show()
```




The boxplots above indicates there are not outliers

In [51]:
```
#Creating patients table
import pandas as pd

data = df.copy()
data.columns = data.columns.str.strip().str.lower().str.replace(" ", "_")

patient_cols = ["name", "age", "gender", "blood_type"]
patients = data[patient_cols].drop_duplicates().reset_index(drop=True)
patients["patient_id"] = "PT-" + (patients.index + 1).astype(str).str.zfill(2)
```

```
# Put patient_id first
patients = patients[["patient_id"] + patient_cols]

patients.head()
```

Out[51]:

| | patient_id | name | age | gender | blood_type |
|---|---|---|---|---|---|
| **0** | PT-01 | Bobby Jackson | 30 | Male | B- |
| **1** | PT-02 | Leslie Terry | 62 | Male | A+ |
| **2** | PT-03 | Danny Smith | 76 | Female | A- |
| **3** | PT-04 | Andrew Watts | 28 | Female | O+ |
| **4** | PT-05 | Adrienne Bell | 43 | Female | AB+ |

In [52]:
```
#Creating admissions table
admissions = data.merge(patients, on=patient_cols, how="left")

admissions["admission_id"] = "AD-" + (admissions.index + 1).astype(str).str.zfill(2

admissions = admissions[[
    "admission_id", "patient_id", "date_of_admission", "discharge_date",
    "admission_type", "hospital", "doctor", "room_number",
    "medical_condition", "medication", "test_results"
]]

admissions.head()
```

Out[52]:

| | admission_id | patient_id | date_of_admission | discharge_date | admission_type | hospital |
|---|---|---|---|---|---|---|
| **0** | AD-01 | PT-01 | 2024-01-31 | 2024-02-02 | Urgent | Sons and Miller |
| **1** | AD-02 | PT-02 | 2019-08-20 | 2019-08-26 | Emergency | Kim Inc |
| **2** | AD-03 | PT-03 | 2022-09-22 | 2022-10-07 | Emergency | Cook PLC |
| **3** | AD-04 | PT-04 | 2020-11-18 | 2020-12-18 | Elective | Hernandez Rogers and Vang, |
| **4** | AD-05 | PT-05 | 2022-09-19 | 2022-10-09 | Urgent | White-White |

In [53]:
```
#Creating billings table
billing = data.copy()

billing["admission_id"] = admissions["admission_id"].values
billing["bill_id"] = "BL-" + (billing.index + 1).astype(str).str.zfill(2)
```

```
billing = billing[[
    "bill_id", "admission_id", "billing_amount", "insurance_provider"
]]

billing.head()
```

Out[53]:

| | bill_id | admission_id | billing_amount | insurance_provider |
|---|---|---|---|---|
| **0** | BL-01 | AD-01 | 18856.281306 | Blue Cross |
| **1** | BL-02 | AD-02 | 33643.327287 | Medicare |
| **2** | BL-03 | AD-03 | 27955.096079 | Aetna |
| **3** | BL-04 | AD-04 | 37909.782410 | Medicare |
| **4** | BL-05 | AD-05 | 14238.317814 | Aetna |

In [54]:
```
#Checks the number of rows in each each table as well as any missing values
print(len(patients), len(admissions), len(billing))
print(admissions["patient_id"].isna().sum(), billing["admission_id"].isna().sum())
```

```
54944 54966 54966
0 0
```

In [55]:
```
!pip install sqlalchemy pymysql
```

```
Requirement already satisfied: sqlalchemy in c:\users\clement\anaconda3\lib\site-pac
kages (2.0.39)
Requirement already satisfied: pymysql in c:\users\clement\anaconda3\lib\site-packag
es (1.2.3)
Requirement already satisfied: greenlet!=0.4.17 in c:\users\clement\anaconda3\lib\si
te-packages (from sqlalchemy) (3.1.1)
Requirement already satisfied: typing-extensions>=4.6.0 in c:\users\clement\anaconda
3\lib\site-packages (from sqlalchemy) (4.12.2)
```

In [59]:
```
# Connecting Python to MySQL database and checks that the connection works by listi
from sqlalchemy import create_engine
from sqlalchemy.engine import URL

url = URL.create(
    "mysql+pymysql",
    username="root",
    password="@Cabenla_13",
    host="localhost",
    port=3306,
    database="hospital_db",
)
engine = create_engine(url)

print(pd.read_sql("SHOW TABLES", engine))
```

```
  Tables_in_hospital_db
0            admissions
1              billings
2              patients
```

In [60]:
```
#This prints the structure of each of the three tables,in order to check that the c
for t in ["patients", "admissions", "billings"]:
    print(t)
    print(pd.read_sql(f"DESCRIBE {t}", engine)[["Field", "Type", "Key"]], "\n")
```

```
patients
        Field         Type  Key
0  patient_id  varchar(10)  PRI
1        name  varchar(30)
2         age          int
3      gender  varchar(10)
4  blood_type   varchar(5)

admissions
                Field          Type  Key
0        admission_id   varchar(10)  PRI
1          patient_id   varchar(10)  MUL
2   date_of_admission          date
3      discharge_date          date
4      admission_type   varchar(30)
5            hospital  varchar(300)
6              doctor   varchar(30)
7         room_number           int
8   medical_condition   varchar(30)
9          medication   varchar(30)
10       test_results   varchar(30)

billings
                Field           Type  Key
0             bill_id    varchar(10)  PRI
1        admission_id    varchar(10)  MUL
2      billing_amount  decimal(12,2)
3  insurance_provider   varchar(100)
```

In [61]:
```
#comparing the column names in MySQL admissions table with the column names in the
print(pd.read_sql("DESCRIBE admissions", engine)["Field"].tolist())
print(admissions.columns.tolist())
```

```
['admission_id', 'patient_id', 'date_of_admission', 'discharge_date', 'admission_typ
e', 'hospital', 'doctor', 'room_number', 'medical_condition', 'medication', 'test_re
sults']
['admission_id', 'patient_id', 'date_of_admission', 'discharge_date', 'admission_typ
e', 'hospital', 'doctor', 'room_number', 'medical_condition', 'medication', 'test_re
sults']
```

In [65]:
```
#Loading the dataframe into MySQL tables
try:
    patients.to_sql("patients", engine, if_exists="append", index=False)
    admissions.to_sql("admissions", engine, if_exists="append", index=False)
    billing.to_sql("billings", engine, if_exists="append", index=False)
    print("Loading done successfully")
except Exception as e:
    print("Loading failed")
    print("Reason:", getattr(e, "orig", e))
```

```
Loading done successfully
```

In [66]:
```
for t in ["patients", "admissions", "billings"]:
    print(t, pd.read_sql(f"SELECT COUNT(*) AS n FROM {t}", engine)["n"][0])

print("DataFrames:", len(patients), len(admissions), len(billing))
```

```
patients 54944
admissions 54966
billings 54966
DataFrames: 54944 54966 54966
```

In [67]:
```
pip install streamlit sqlalchemy pymysql pandas
```

```
Requirement already satisfied: streamlit in c:\users\clement\anaconda3\lib\site-pack
ages (1.45.1)Note: you may need to restart the kernel to use updated packages.

Requirement already satisfied: sqlalchemy in c:\users\clement\anaconda3\lib\site-pac
kages (2.0.39)
Requirement already satisfied: pymysql in c:\users\clement\anaconda3\lib\site-packag
es (1.2.3)
Requirement already satisfied: pandas in c:\users\clement\anaconda3\lib\site-package
s (2.2.3)
Requirement already satisfied: altair<6,>=4.0 in c:\users\clement\anaconda3\lib\site
-packages (from streamlit) (5.5.0)
Requirement already satisfied: blinker<2,>=1.5.0 in c:\users\clement\anaconda3\lib\s
ite-packages (from streamlit) (1.9.0)
Requirement already satisfied: cachetools<6,>=4.0 in c:\users\clement\anaconda3\lib
\site-packages (from streamlit) (5.5.1)
Requirement already satisfied: click<9,>=7.0 in c:\users\clement\anaconda3\lib\site-
packages (from streamlit) (8.1.8)
Requirement already satisfied: numpy<3,>=1.23 in c:\users\clement\anaconda3\lib\site
-packages (from streamlit) (2.1.3)
Requirement already satisfied: packaging<25,>=20 in c:\users\clement\anaconda3\lib\s
ite-packages (from streamlit) (24.2)
Requirement already satisfied: pillow<12,>=7.1.0 in c:\users\clement\anaconda3\lib\s
ite-packages (from streamlit) (11.1.0)
Requirement already satisfied: protobuf<7,>=3.20 in c:\users\clement\anaconda3\lib\s
ite-packages (from streamlit) (6.33.6)
Requirement already satisfied: pyarrow>=7.0 in c:\users\clement\anaconda3\lib\site-p
ackages (from streamlit) (19.0.0)
Requirement already satisfied: requests<3,>=2.27 in c:\users\clement\anaconda3\lib\s
ite-packages (from streamlit) (2.32.3)
Requirement already satisfied: tenacity<10,>=8.1.0 in c:\users\clement\anaconda3\lib
\site-packages (from streamlit) (9.0.0)
Requirement already satisfied: toml<2,>=0.10.1 in c:\users\clement\anaconda3\lib\sit
e-packages (from streamlit) (0.10.2)
Requirement already satisfied: typing-extensions<5,>=4.4.0 in c:\users\clement\anaco
nda3\lib\site-packages (from streamlit) (4.12.2)
Requirement already satisfied: watchdog<7,>=2.1.5 in c:\users\clement\anaconda3\lib
\site-packages (from streamlit) (4.0.2)
Requirement already satisfied: gitpython!=3.1.19,<4,>=3.0.7 in c:\users\clement\anac
onda3\lib\site-packages (from streamlit) (3.1.43)
Requirement already satisfied: tornado<7,>=6.0.3 in c:\users\clement\anaconda3\lib\s
ite-packages (from streamlit) (6.5.1)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\clement\anaconda3
\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\clement\anaconda3\lib\site-p
ackages (from pandas) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\clement\anaconda3\lib\site
-packages (from pandas) (2025.2)
Requirement already satisfied: jinja2 in c:\users\clement\anaconda3\lib\site-package
s (from altair<6,>=4.0->streamlit) (3.1.6)
Requirement already satisfied: jsonschema>=3.0 in c:\users\clement\anaconda3\lib\sit
e-packages (from altair<6,>=4.0->streamlit) (4.23.0)
Requirement already satisfied: narwhals>=1.14.2 in c:\users\clement\anaconda3\lib\si
te-packages (from altair<6,>=4.0->streamlit) (1.31.0)
Requirement already satisfied: colorama in c:\users\clement\anaconda3\lib\site-packa
ges (from click<9,>=7.0->streamlit) (0.4.6)
Requirement already satisfied: gitdb<5,>=4.0.1 in c:\users\clement\anaconda3\lib\sit
```

```
e-packages (from gitpython!=3.1.19,<4,>=3.0.7->streamlit) (4.0.7)
Requirement already satisfied: smmap<5,>=3.0.1 in c:\users\clement\anaconda3\lib\sit
e-packages (from gitdb<5,>=4.0.1->gitpython!=3.1.19,<4,>=3.0.7->streamlit) (4.0.0)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\clement\anaconda
3\lib\site-packages (from requests<3,>=2.27->streamlit) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in c:\users\clement\anaconda3\lib\site-p
ackages (from requests<3,>=2.27->streamlit) (3.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\clement\anaconda3\lib
\site-packages (from requests<3,>=2.27->streamlit) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\clement\anaconda3\lib
\site-packages (from requests<3,>=2.27->streamlit) (2025.8.3)
Requirement already satisfied: greenlet!=0.4.17 in c:\users\clement\anaconda3\lib\si
te-packages (from sqlalchemy) (3.1.1)
Requirement already satisfied: attrs>=22.2.0 in c:\users\clement\anaconda3\lib\site-
packages (from jsonschema>=3.0->altair<6,>=4.0->streamlit) (24.3.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in c:\users\clem
ent\anaconda3\lib\site-packages (from jsonschema>=3.0->altair<6,>=4.0->streamlit) (2
023.7.1)
Requirement already satisfied: referencing>=0.28.4 in c:\users\clement\anaconda3\lib
\site-packages (from jsonschema>=3.0->altair<6,>=4.0->streamlit) (0.30.2)
Requirement already satisfied: rpds-py>=0.7.1 in c:\users\clement\anaconda3\lib\site
-packages (from jsonschema>=3.0->altair<6,>=4.0->streamlit) (0.22.3)
Requirement already satisfied: six>=1.5 in c:\users\clement\anaconda3\lib\site-packa
ges (from python-dateutil>=2.8.2->pandas) (1.17.0)
Requirement already satisfied: MarkupSafe>=2.0 in c:\users\clement\anaconda3\lib\sit
e-packages (from jinja2->altair<6,>=4.0->streamlit) (3.0.2)
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